BLK II - UNIT 2: Basic Forecast Techniques
b. Basic Forecasting Techniques

The goal of this objective will be to show you have the capability to use meteorological reasoning to produce a basic forecast of various weather features. To successfully demonstrate competency on this topic, you will be given information through lecture, a series of questions and tasks designed to display your ability to correctly predict the future state of the atmosphere using the tools discussed throughout this objective. These foundational forecasting concepts will be measured with a Progress Check (PC) at the end of the objective and must be passed to continue onto the next objective. Follow your instructor's guidance as they provide information and instruction for completing all requirements necessary to pass this objective.

Cloud Forecast Techniques

Clouds form when water vapor changes to either liquid droplets or ice crystals. This requires an atmospheric cooling process. Adiabatic cooling from free or forced convection is the most effective cooling process.

Convection and stability determine the type of cloud and precipitation. An unstable atmosphere (cold over warm temperatures) will develop cumuliform clouds, while a stable atmosphere (warm over cold temperatures) will develop stratiform clouds. available moisture is more important than stability for cloud development.

Dewpoint Depression Method

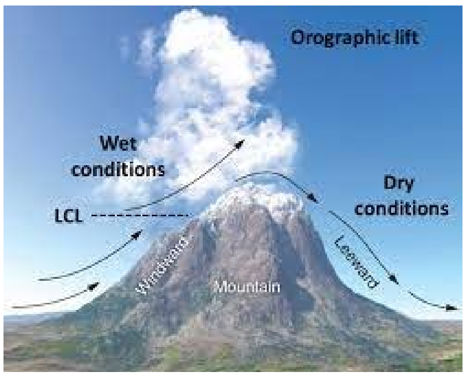
Forecast the height of cumulus cloud bases by inserting the current or forecast surface dewpoint depression into Table 2. This table is not suitable for use at stations situated in mountainous or hilly terrain and should be used only when clouds are formed by active surface convection in the vicinity of the station.

Base of Convective Clouds using Surface Dewpoint Depressions			
Dewpoint Depression (°C)	Estimated Cumulus Height (ft)	Dewpoint Depression (°C)	Estimated Cumulus Height (ft)
0.5	200	1.0	400
1.5	600	2.0	800
2.5	1,000	3.0	1,200
3.5	1,400	4.0	1,600
4.5	1,800	5.0	2,000
5.5	2,200	6.0	2,400
6.5	2,600	7.0	2,800
7.5	3,000	8.0	3,200
8.5	3,400	9.0	3,600
9.5	3,800	10.0	4,000
10.5	4,200	11.0	4,400
11.5	4,600	12.0	4,800
12.5	5,000		

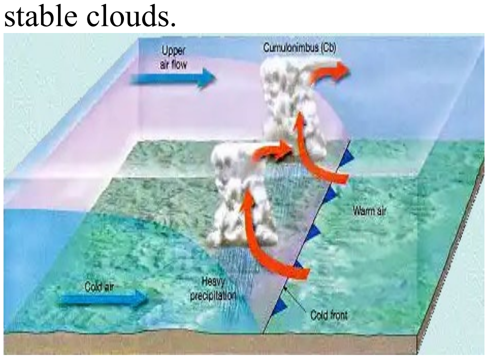
Table 2: Base of convective clouds using surface dewpoint depressions.

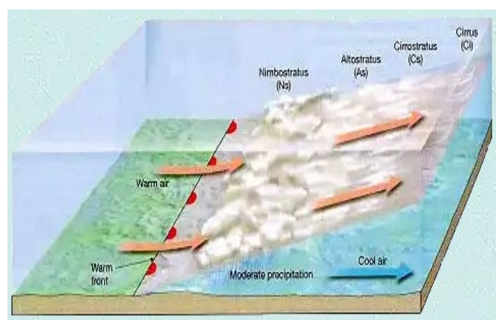
Lifting Mechanism
There are four main types of lifting mechanisms: orographic, frontal, low-level convergent flow, and surface heating.

Orographic: Orographic lift is the most effective and intense cooling process. The overall effect is dependent on the change from horizontal surface flow to vertical motion proportional to the terrain's slope. The strongest or greatest amount of lift occurs with a steep slope, while a gradual or relatively flat slope has the weakest amount of orographic lift.

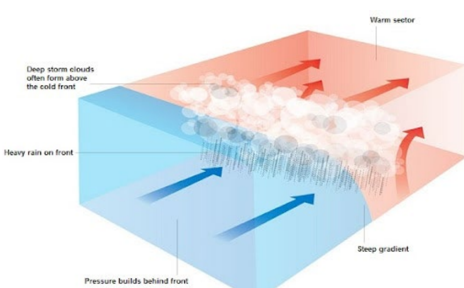


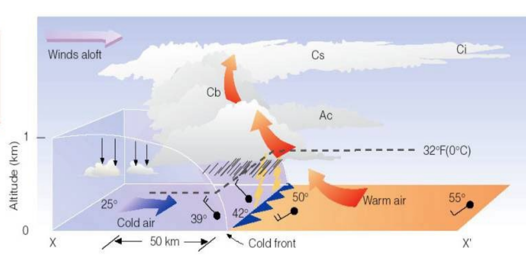
Frontal: Frontal lift occurs with all fronts. Steep slopes (cold fronts) produce convective cloud types, while a shallow slope (warm and stationary fronts) produce widespread stable clouds.



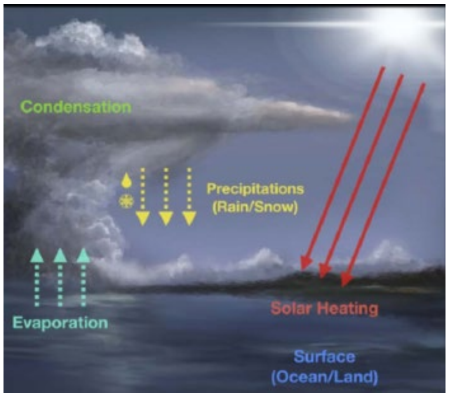


Low-Level Confluent Flow: Low-level confluent flow is common in the warm sector of a baroclinic low ahead of an approaching cold front. Enhanced cumuliform clouds develop when low-level confluent flow combines with diffluent upper-level flow.





Surface Heating: Surface heating produces convective currents which may produce cumuliform clouds. Cloud formation may be random and isolated at inland locations, or it may organize into a linear form along a sea or land breeze.



Stratiform

Stratiform clouds form within stable atmospheric layers.

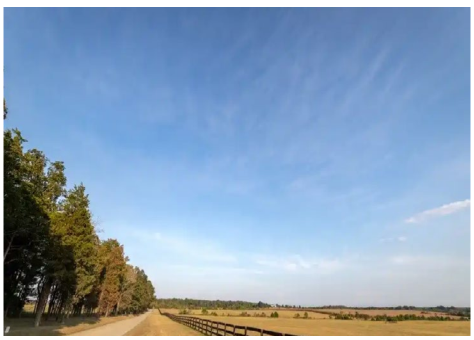
Stratus (ST). There must be cooling in the surface moist layer for stratus clouds to form. They are typical with warm air masses and are normally 1,000 to 4,000 feet thick with basses less than 2,000 feet above ground level (AGL). Conditions favorable for development include:

- Warm frontal zones where warm, moist air are advected over a cooler surface. Clouds form near the front due to slight upward vertical motion.
- Along gradual upslope conditions due to orographic lift.
- Areas of weak surface confluence with high moisture content.

Stratus clouds dissipate by removing moisture from the atmosphere. Thin layers typically do not take long to dissipate, but thicker layers take longer since they inhibit insolation (surface heating). Warmer surface temperatures can absorb the moisture thereby dissipating the stratus.

Altostratus (AS). Altostratus clouds are associated with wave cyclones. They are poleward of the warm front with unstable waves and poleward of the warm front as well as along the western edge of a comma cloud.

Cirrostratus (CS). Cirrostratus forms with well-developed wave cyclones normally 700 to 1,000 miles ahead of a warm front. The clouds may move with the gradient flow days after the frontal system dissipates.

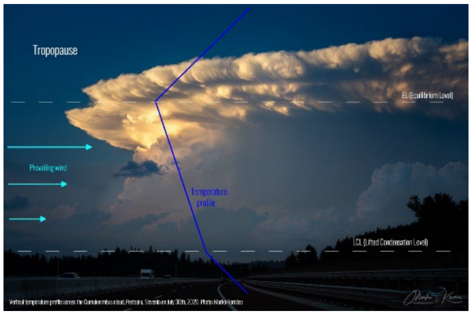


Cirrus (CI). Cirrus is often associated with the remnants of a cumulonimbus cloud and can remain in the atmosphere after the storm collapses.

Cumuliform Clouds

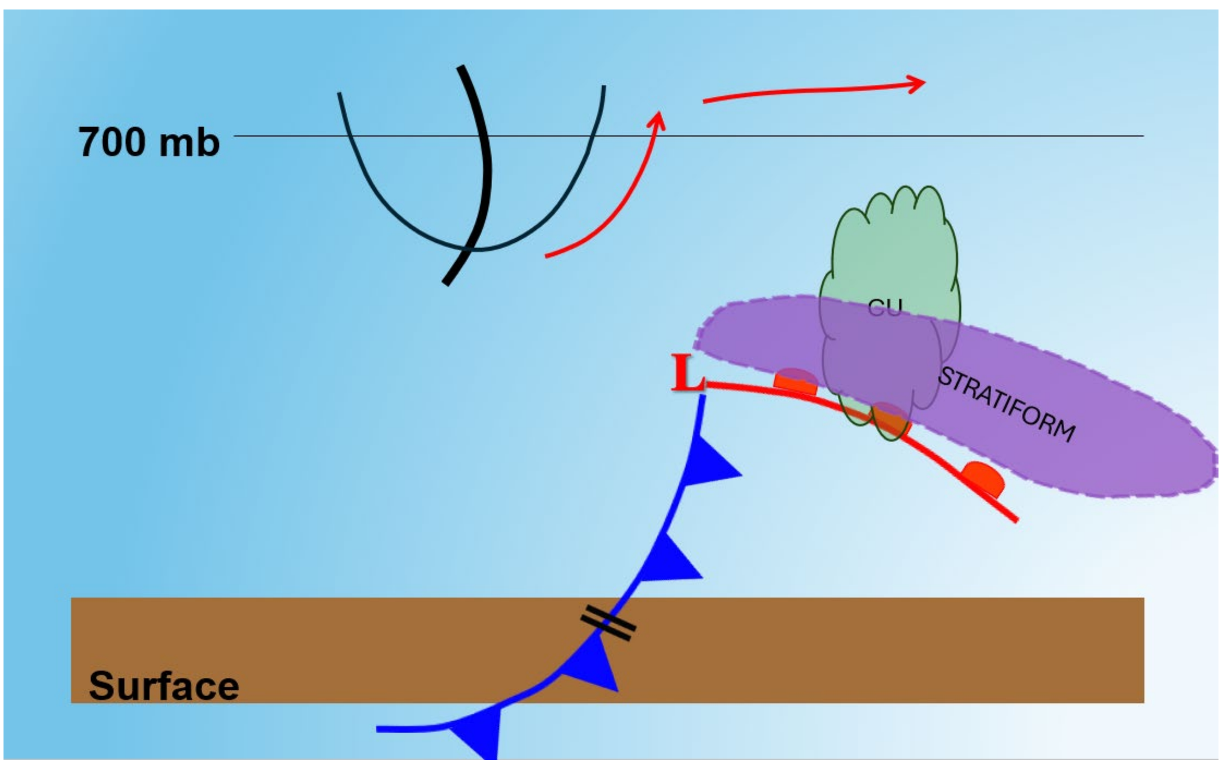
Cumuliform clouds form in unstable environments. The cloud types include cumulus, towering cumulus, and cumulonimbus. Their bases vary based on the lifting mechanism. When moist air rises mechanically by frontal or orographic lift, cloud bases form near the lifted condensation level (LCL). When the air rises thermally from surface heating, cloud bases are closer to the convective condensation level (CCL).

Cumuliform tops are dependent on the height of the instability layer. Their vertical development ends at the equilibrium layer (EL). This is where air parcels' temperature and pressure equal the environment therefore causing vertical motion to stop. The equilibrium layer may be a stable isothermal layer, an inversion, or the tropopause.

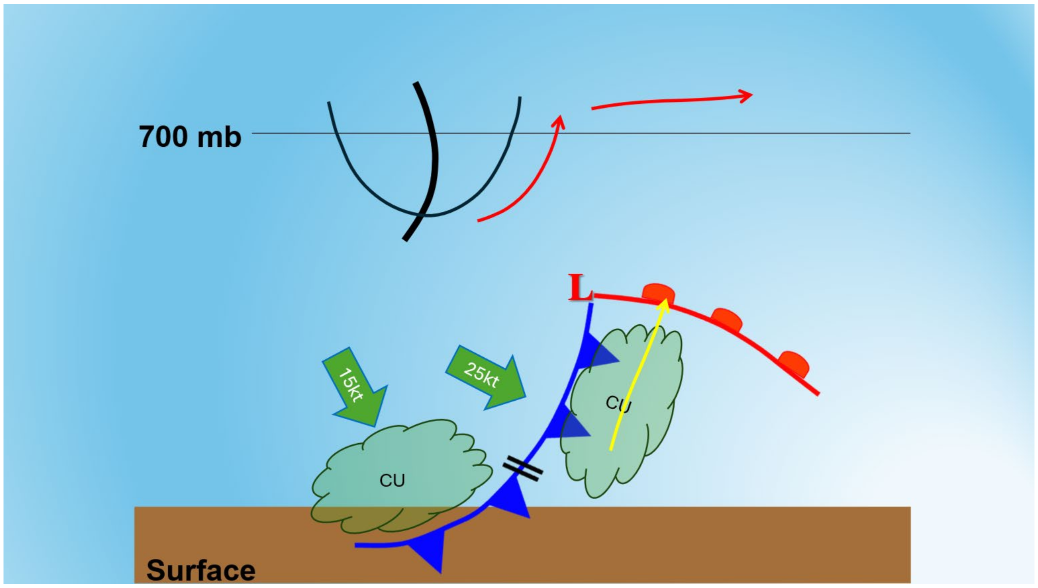


Frontal Clouds

Warm Fronts: Warm frontal clouds are generally stratiform unless the atmosphere is unstable. Embedded cumulus may occur in those instances. Another characteristic of warm frontal clouds is based on the 700-mb flow over the frontal boundary. When the flow is cyclonic or straight-line, embedded cumulus may form and when it is anticyclonic, there is less potential for clouds and precipitation poleward of the frontal boundary.



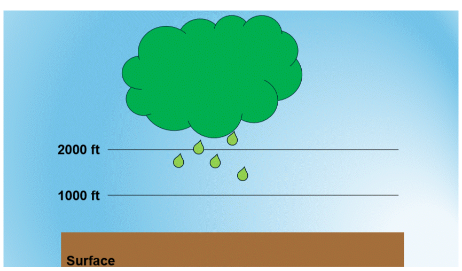
Inactive Cold Fronts: Most clouds form on the leading edge (downstream side) of inactive cold fronts perpendicular to the 700 mb flow. The front has the steepest slope and moves the fastest at an average speed of 25 knots causing rapid clearing of all clouds after frontal passage.



Active Cold Fronts: Most clouds form at and behind (upstream of) active cold fronts. The cloud shield is more extensive since the frontal slope is lower and its speed (average 15 knots) is slower than an inactive cold front. The overall result is clouds clear slowly after frontal passage.

Cloud Bases and Precipitation

Precipitation lowers cloud bases. Continuous precipitation of sufficient duration can also lower cloud bases below 2,000 feet often by 800 feet. However, the actual height of the cloud base can vary depending on the location, terrain, and latitudinal differences. Showery or heavy precipitation can also rapidly drop cloud bases below 2,000 feet.

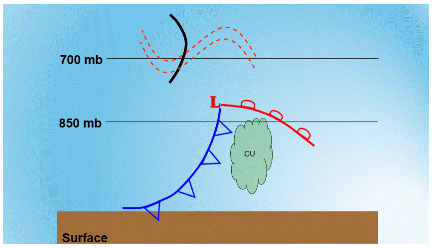


Clouds Dissipation

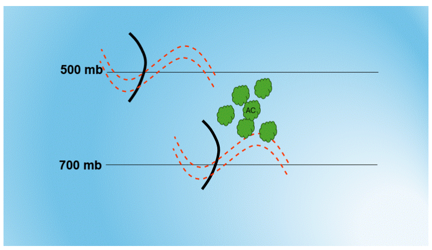
Clouds dissipate by removing moisture through dry air advection, adiabatic warming (downward vertical motion), or removing the lifting mechanism.

Cold Frontal Clouds: Cloud layers will dissipate or move with the frontal system in the following order of events:

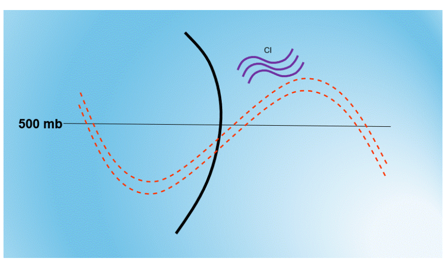
- Low étage clouds begin to move and dissipate with passage of the 850-mb upper front and they completely move away with passage of the 700-mb frontal trough.



- Middle étage clouds begin to move and dissipate with passage of the 700-mb frontal trough and they completely move away with passage of the 500-mb frontal trough.



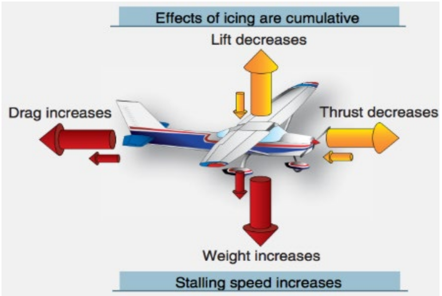
- High étage clouds begin to move and dissipate with passage of the 500-mb frontal trough and they completely move away with passage of the 500-mb thermal trough.



Icing Forecast Techniques

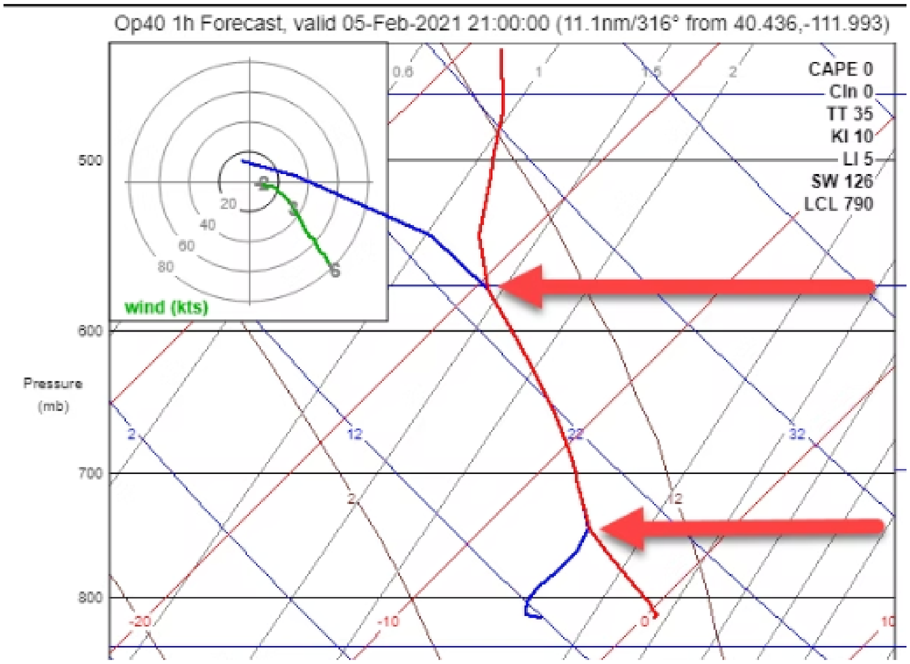
Effects of Aircraft Icing

Icing increases drag and weight of an aircraft, while decreasing its lift capabilities. Other structural effects include loss of aircrew's vision, instrument failure, and loss of radio communication. Engine-system icing reduces the effective power of aircraft engines, potentially causing them to stall and stop working.



Formation Requirements

Icing formation requires suspended super-cooled water droplets with temperatures below freezing and moisture available. Regarding temperature, most icing forms between 0°C and -22°C. Supercooled water droplets can form between -23°C and -40°C, however, droplets with temperatures less than -40°C are completely frozen. So, icing does not exist with temperatures below -40°C.



Severity

There are definite criteria defining a pilot's reactions for icing during flight operations. Be aware of the icing effects differ between aircraft type. The differences include fixed versus rotary wing aircraft, their operational flight altitudes, and available de-icing equipment. For example, de-icing equipment is not available on all military aircraft. To differentiate between the two, deicing equipment melts ice on the aircraft during flight and anti-icing procedures prevents ice accumulation which is conducted on the runway prior to takeoff.

Trace

The first level occurs when ice becomes perceptible meaning the pilot recognizes it is on the aircraft. De-icing equipment or anti-icing procedures are not necessary. Trace icing is not hazardous unless the aircraft encounters the icing for more than one hour.

Light

Light icing occurs when ice accumulates on the aircraft. The rate of accumulation is slow and not a hazard unless the aircraft is in the condition for more than one hour. Occasional use of deicing equipment or anti-icing procedures are required.

Moderate
The rate of ice accumulation with a moderate intensity is quick making short encounters hazardous to aircraft operations. De-icing equipment or anti-icing procedures are necessary for flight safety and diversion may be required if the de-icing equipment fails.

Severe
An environment with severe icing occurs when de-icing equipment does not reduce ice accumulation. Immediate diversion is necessary to continue aircraft operations.

Types
Clear Icing
Clear icing forms when super-cooled water droplets gradually freeze when striking an aircraft. This is usually associated with precipitation since it begins with large, super-cooled water droplets. The unfrozen water droplet flows or smears over the airframe and freezes. The ice accumulates on the aircraft as a thin, smooth surface. Clear icing is hard and heavy making it very difficult to remove.

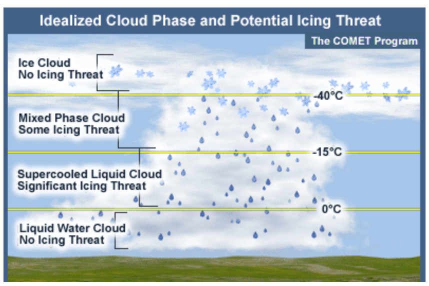
Rime Icing
Rime icing forms by the instantaneous freezing of small, super- cooled water droplets which traps large amounts of air within the ice. It is associated with super-cooled cold droplets within warm clouds. Rime icing has a milky, opaque, and granular appearance. It is also lightweight and brittle making it easy to remove.

Mixed Icing
Mixed icing is a combination of rime and clear icing. It is formed most often where cloud and precipitation droplets intermix. These areas have water droplets varying in liquid water content, droplet size, and temperature. It appears as layers of clear and opaque ice due to ice particles embedding into the smoother clear ice as it freezes on the aircraft. Overall, it has a rough appearance in thin layers with embedded air pockets.

Icing and Cloud Type

Stratiform Clouds
Low and middle étage stratiform clouds are associated with mixed and rime icing. Their intensity ranges from light to moderate.

Cumuliform Clouds
Clear or mixed icing is always present in the updrafts of a cumulonimbus cloud. Light icing occurs with small cumulus clouds and moderate icing is within towering cumulus clouds. Severe icing is always associated with cumulonimbus clouds.



Cirriform Clouds
Icing rarely occurs with cirriform clouds. The exception is the dense cirrus and anvil tops of a cumulonimbus cloud where there is a considerable amount of water in the updraft.

Frontal Systems

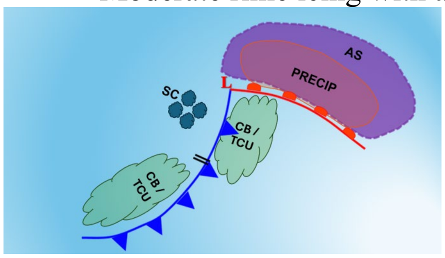
85% of reported icing conditions occur with frontal systems with the greatest amount associated with warm and cold fronts.

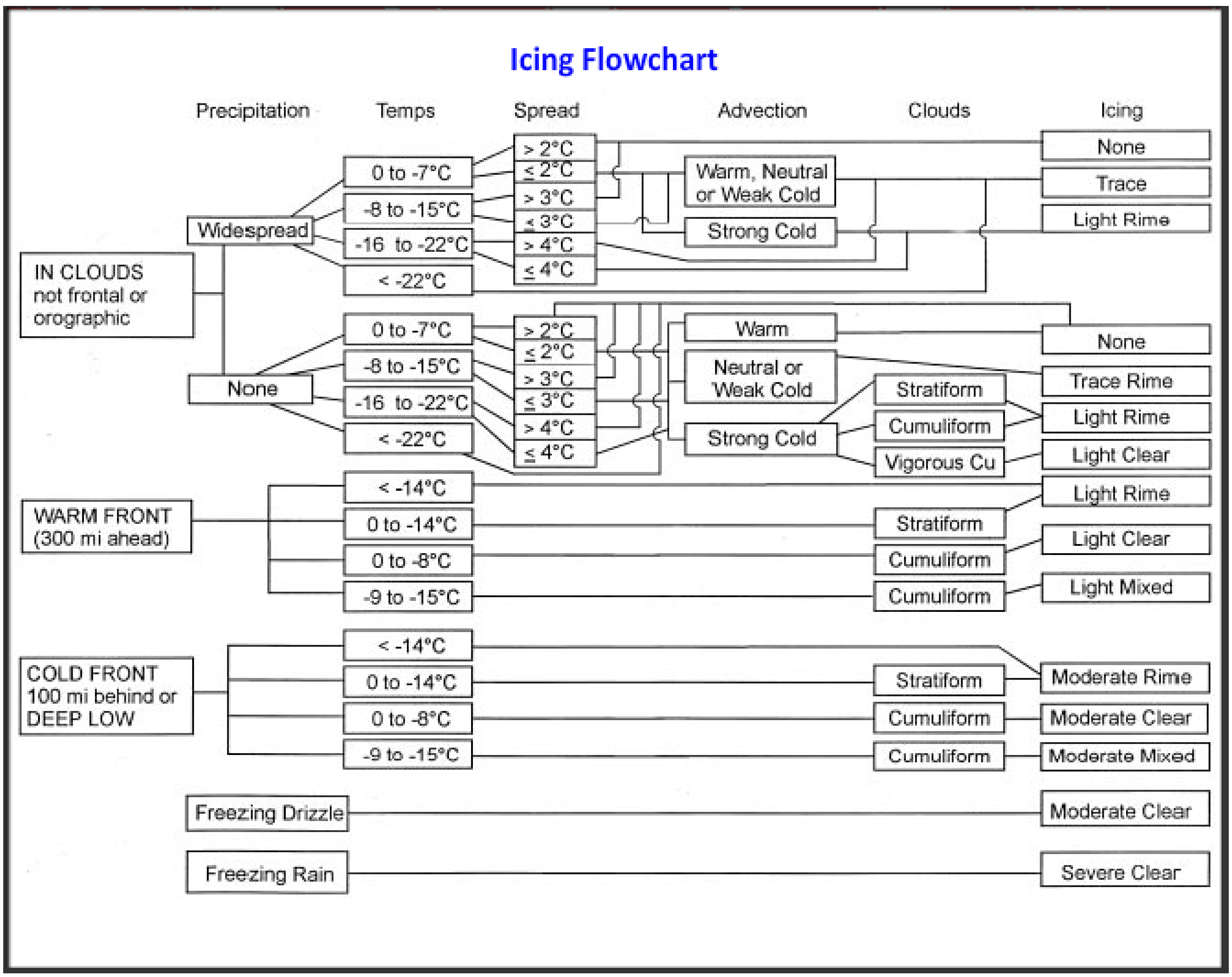
Warm Fronts
Icing forms on the poleward side of a warm front in a widespread area. Light rime icing occurs in altostratus clouds up to 300 miles ahead of the warm front surface. Light mixed icing occurs along the frontal slope where clouds and precipitation mix. The icing layer may extend 100 to 200 miles downstream of the front. Lastly, clear icing occurs in precipitation and cold air below the warm frontal slope. Intensities can vary based on the precipitation type.

Cold Fronts
Icing forms in narrow regions with cold fronts due to less extensive cloud coverage. Stratocumulus clouds behind the cold front have light mixed icing when precipitation is present or light rime when there is no precipitation. Icing also develops in cumulonimbus and towering cumulus clouds up to 100 miles from the frontal boundary. The area may be upstream of an

active cold front or downstream of an inactive cold front. Regardless of location, there are three potential icing combinations.

- Moderate clear icing with average temperatures between 0 and -08°C
- Moderate mixed icing with average temperatures between -09 and -15°C
- Moderate rime icing with average temperatures between -16 and -22°C.

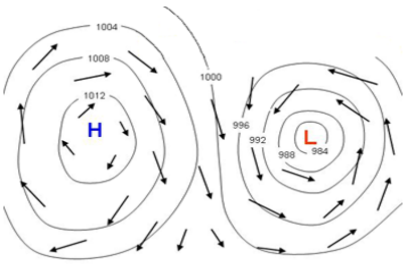




Exercise / Practice: Clouds & Icing Forecast

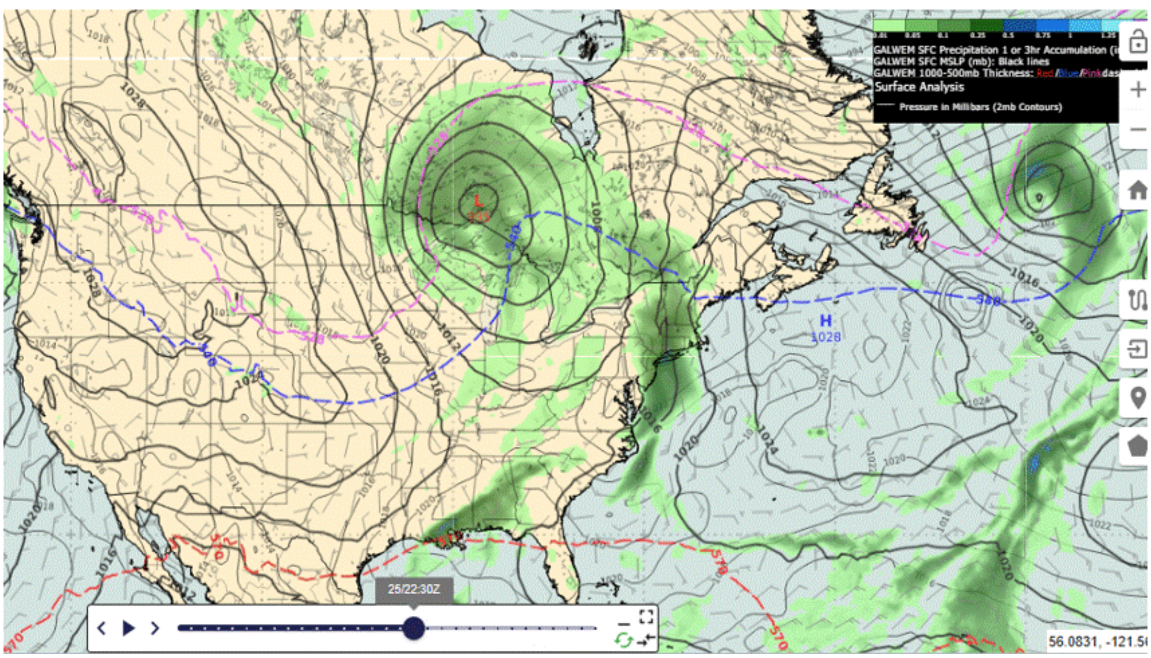
Wind Forecast Techniques

Wind flows from high to low pressure. High-pressure centers have anticyclonic flow and a weak isobaric gradient resulting in slower surface wind speeds. Low-pressure centers have cyclonic flow and a tighter isobaric gradient resulting in faster wind speeds.



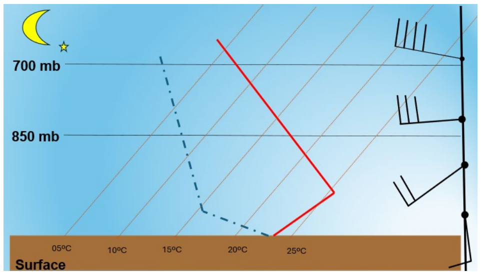
Synoptic Winds

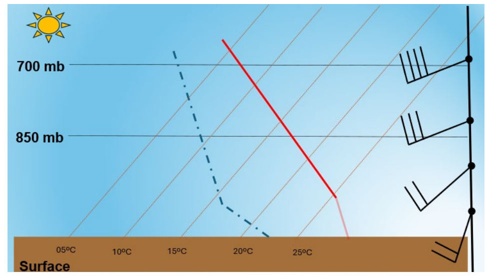
Synoptic winds stay consistent with the pressure system's forecast meaning they are directly proportional to the approaching pressure center. High pressure centers cause wind direction to veer anticyclonically, while low pressure centers cause wind direction to back cyclonically. Speed is proportional to the pressure gradient. A loose or decreasing isobaric gradient results in slower speeds, while a tight or increasing isobaric gradient results in faster speeds.



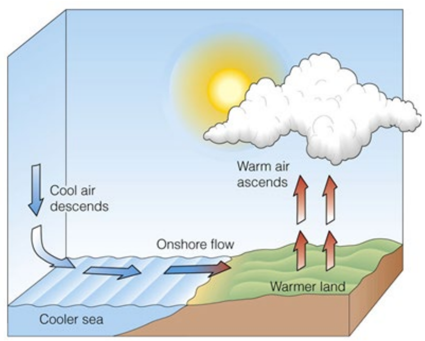
Diurnal Trends

Consider diurnal trends in every forecast. If the air mass does not change, then the winds should be nearly the same as the previous day's wind direction and speed. They decrease at night due to atmospheric cooling and formation of a radiation inversion. Surface winds have low speeds below the radiation inversion and stronger winds above the radiation inversion. The lightest surface winds occur near sunrise when the inversion is the strongest. Daytime heating initiates free convection after sunrise and breaks the inversions by mid-morning resulting in higher wind speeds.

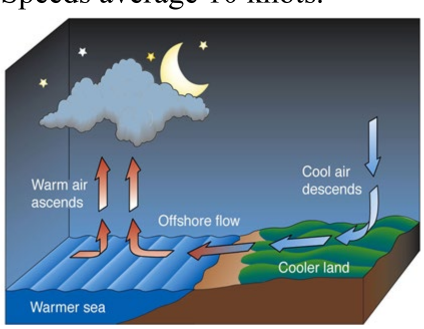




Sea Breeze
Sea breezes occur during the day. They form when cooler air over water moves over warmer land with a weak pressure gradient. They begin to develop 3 to 4 hours after sunrise and often extends 10 miles inland but may extend as far as 30-40 miles inland. Speeds average between 10 and 20 knots.



Land Breeze
Land breezes occur during the night. They form when cooler air over land moves over warmer water with a weak pressure gradient. They develop after midnight and reach their maximum speed slightly before sunrise. Forecast wind direction offshore perpendicular to the coastline. Speeds average 10 knots.



Upslope Winds
Upslope winds occur during the day or night. They form when the isobaric pattern is perpendicular to a mountain or mountain range. Forecast wind speeds are based on the pressure gradient. Tighter gradients produce strong wind speeds and possible upslope clouds. Weaker gradients result in light wind speeds with upslope stratus. Upslope winds cease or never form when the synoptic isobaric pattern is parallel to the mountain range.

Downslope Winds
Downslope winds form with a nearly perpendicular synoptic contour pattern along the mountain range in the upper levels. Forecast wind direction based on the contour gradient crossing the mountain tops. Strong wind speeds are near mountain passes and downstream parallel to the surface isobaric gradient. Speeds can reach 100 knots when channeled (funneled).

Turbulence Forecast Techniques
Turbulence is any irregular motion of air such as random and continuously changing air direction and speeds within the synoptic flow pattern. Turbulence with fixed wing aircraft is directly proportional to aircraft speed and wing area but inversely proportionate to weight. Turbulence on rotary wing aircraft are directly proportional to their speed and rotary blades but inversely proportionate to weight and lift velocities.


Categories

There are six turbulence categories. They are:

- Light turbulence - slight erratic changes in altitude and/or attitude (stability at flight level)
- Light chop - slight, rapid rhythmic bumpiness without any appreciable changes in altitude and/or attitude.
- Moderate turbulence - small changes in altitude and/or attitude but the pilot always stays in positive control of the aircraft. Usually indicated by variations in air speed.
- Moderate chop - rapid bumps or jolts without changes in altitude and/or attitude.
- Severe turbulence - large changes in altitude and/or attitude and aircraft is difficult to control. Usually indicated by large variations in air speed,
- Extreme turbulence - large sudden changes in altitude and/or attitude. Aircraft is violently tossed around and virtually impossible to control. Structural damage may occur.

Turbulence Types

Convective (Thermal). Convective turbulence occurs where isolated vertical air currents form due to surface heating. We know warm air rises and cold air sinks. When these are next to each other they produce bumpy conditions, pilots refer to as CHOP. Convective turbulence affects the lower atmosphere typically from surface to 10,000 feet. It occurs most often in the late morning through late afternoon. Favorable conditions for its formation include warm summer afternoons with clear to partly cloudy skies, light surface winds, and uneven terrain with different types of surfaces. Its intensity is normally light to moderate but can be moderate to severe when associated with thunderstorms.

Convective

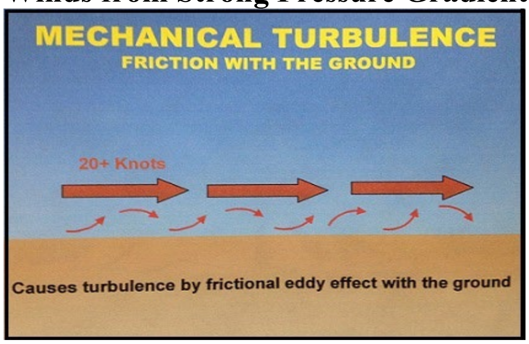
Mechanical Turbulence. Wind flowing over irregular terrain or through and around obstacles creates mechanical turbulence. Its intensity depends on a variety of factors:

- Roughness of terrain
- Height, configuration, and specific density of obstructions
- Wind speeds
- Air mass stability.

The mechanical turbulence layer is usually 2,000 feet thick, 10 to 40 miles wide, and can extend hundreds of miles long. Actual numbers will vary, but mechanical turbulence areas are always longer than they are wide. Favorable conditions require a strong horizontal pressure gradient, irregular and mountainous terrain, frontal zones, or radiation inversions.

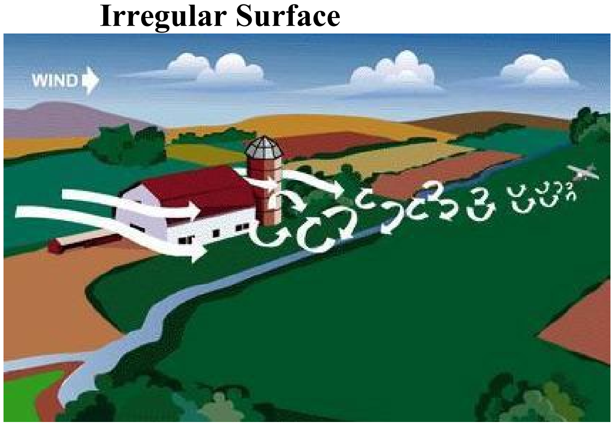
- Strong pressure gradients create stronger wind speeds which increases turbulent mixing near the surface creating mechanical turbulence.

Winds from Strong Pressure Gradient



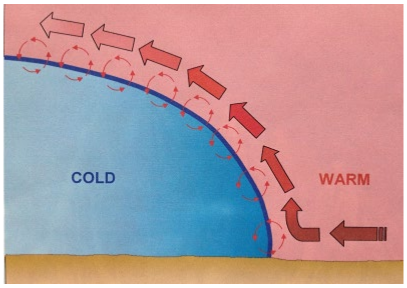
- Irregular or mountainous terrain, especially with greater terrain irregularity or steeper slopes produces higher turbulence intensities due to a grater vertical extent of the turbulent mixing.

Irregular Surface



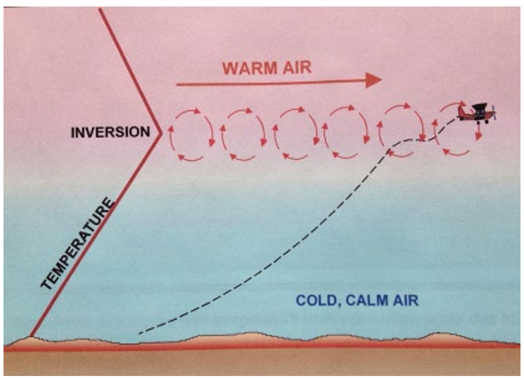
- Warm sector winds force air aloft along the frontal slopes of an upstream cold front and downstream of warm fronts.

Frontal



- Larger currents occur with inactive cold fronts and smaller currents occur with warm fronts.
- Radiation inversions have lighter surface winds trapped below the inversion, while stronger wind speeds are aloft above the inversion. Mechanical turbulence forms along the frontal zones or the radiation inversion's boundaries.

Radiation Inversion

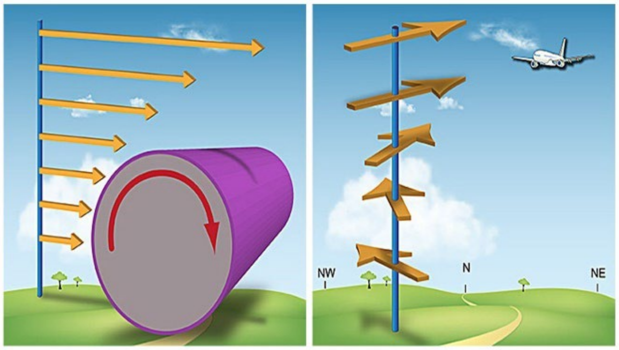


Low-level Wind Shear (LLWS). LLWS is a form of low-level mechanical turbulence. It has a specific definition which makes it a separate turbulence type. LLWS occurs with abrupt changes in wind speed and/or direction within 2,000 feet of the surface associated with airspeed changes greater than 15 knots and/or vertical speed changes greater than 500 feet per minute. It is extremely dangerous for fixed and rotary wing aircraft during take-off and landing as well as during flights below 2,000 feet AGL. These operations are too close to the surface so if the aircraft experiences LLWS, the pilot may be unable to react appropriately and may result in crashing into the aircraft. Locations of LLWS include:

- Near Thunderstorms: This exists horizontally along outflow boundaries and can extend 10 to 20 miles from the thunderstorm. It also exists vertically between updrafts and downdrafts in thunderstorms.
- Frontal Boundaries: The strongest intensities will occur with an inactive cold front from frontal passage and up to 2 hours behind the cold front. With warm fronts, is weaker than cold fronts but the wind shear area extends up to 6 hours ahead of a warm frontal passage.
- Low-Level Jet: Forms with a low-level jet above a radiation inversion. In this instance, winds below the inversion are typically less than 10 knots and winds above the inversion

are 30 knots or more. Speed changes and vertical distance determines the intensity of LLWS.

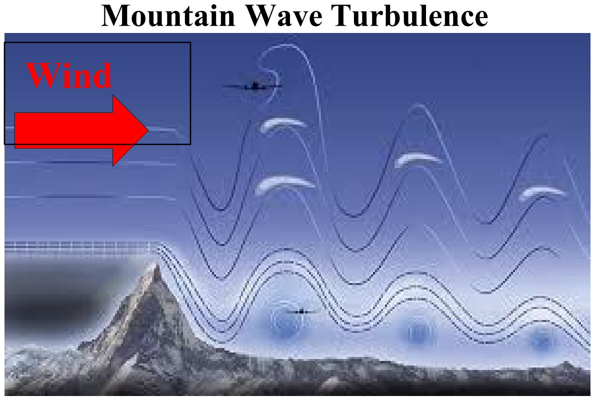
Low Level Wind Shear



Mountain Wave Turbulence. Mountain wave turbulence extends downstream of mountain ranges. It is commonly associated with higher mountains but can exist in any mountain range. It can extend as high as 70,000 feet and typically extends as far as 300 miles downstream from the mountain range. The strongest turbulent areas occur with a strong perpendicular wind component to the mountain range. Favorable conditions for formation include a stable layer at and above the mountain tops, a subsidence inversion compressing the atmosphere vertically, and downward vertical motion combining with the perpendicular wind component.

- If the perpendicular wind component within 2,000 feet of the mountain tops and the wind speed is 25-49 knots, forecast severe turbulence up to 50 miles downstream and moderate turbulence 50 to 150 miles downstream of the mountain range.
- When the perpendicular wind component is 50 knots or more, forecast extreme turbulence up to 10 miles, severe turbulence between 10 and 150 miles, and moderate turbulence from 150 to 300 miles downstream of the mountain range.

Mountain Wave Turbulence

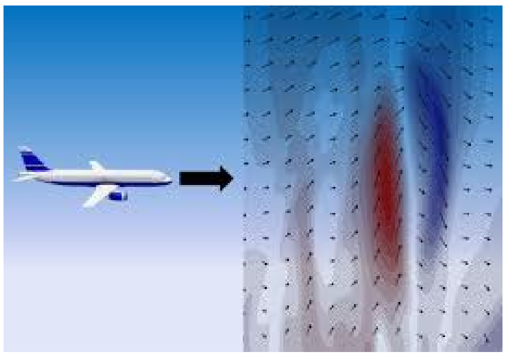


Clear Air Turbulence (CAT). Clear air turbulence forms in clear air. It is typically an elongated area of strong winds forming more frequently during the winter. Atmospheric conditions favorable for CAT are:

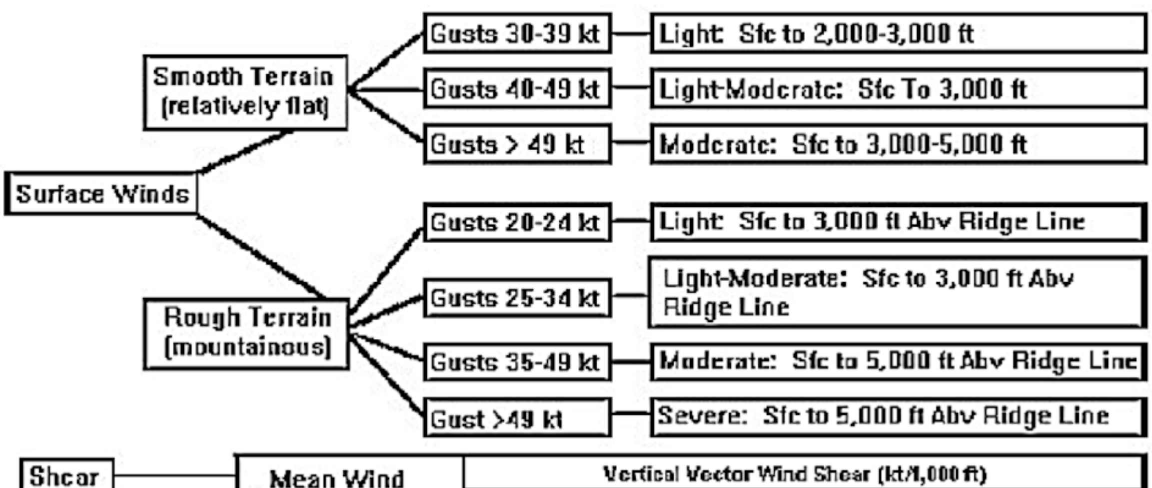
- Anticyclonic oriented jet stream. Forecast moderate CAT symmetrical to the jet axis when it flows around a medium to high-amplitude long-wave ridge axis. The turbulent area extends 250 miles on both sides of the ridge axis.

- Areas of deep troughs or sharp ridges. Forecast moderate to severe CAT when winds exceed 135 knots with large curvature or winds exceed 115 knots with sharp curvature.
- Developing cut-off lows. Forecast moderate CAT in the opposing winds and subsequent wind shear zones associated with the confluent or diffluent contours poleward of a low-height center or 500-mb cold pocket.
- Closed surface lows undergoing cyclogenesis. CAT forms near the jet core height north to northeast of a developing surface low-pressure center. Turbulence intensity is based on the rate of cyclogenesis.

Clear Air Turbulence



Turbulence Flowchart



Mean Wind Speed in Layer	Vertical Vector Wind Shear (kt/1,000 ft)					
	0 - 8	8 - 10	11 - 20	21 - 30	31 - 50	>50
40 - 60 kt	N	L	L - M	M	M - S	S
61 - 120 kt	L	L - M	M	M - S	S	S - X
>120 kt	L	L - M	M	M - S	S	S - X

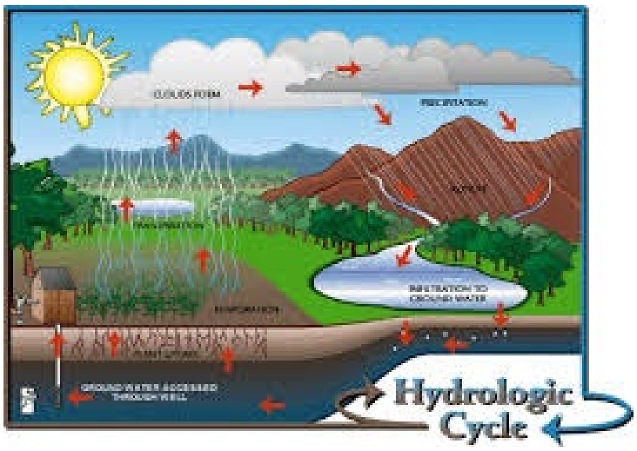
N = None; L = Light; M = Moderate; S = Severe; X = Extreme

Exercise / Practice: Wind & Turbulence Forecast

Precipitation Forecast Techniques

Formation Requirements

Precipitation formation requirements include ample moisture, and instability. The process begins by forming a cloud near or moving towards an ample moisture source. Increased moist air advection feeds the cloud especially if the synoptic flow becomes more perpendicular from the moisture source to the cloud's position. Increasing instability is also a crucial factor whether it is from a lifting mechanism such as surface convergence or divergence aloft.



Precipitation-cloud Relationships

Precipitation cloud relationships are complex and different types of clouds can produce different types of precipitation. A quick synopsis of each includes:

- Cumulonimbus clouds are associated with showery precipitation that may include rain, snow, or ice pellets, and hail.
- Cumulus clouds must reach a vertical development of 8,000 to 10,000 feet or more with the cloud tops reaching heights of -12 to -20°C or colder. Precipitation types may be rain, snow, or ice pellets.
- Stratocumulus clouds typically produce light rain, snow, or snow pellets. If the cloud base is low drizzle can occur.
- Stratus clouds may produce light to heavy drizzle, rain, snow, and snow grains.
- Altostratus clouds produce continuous rain, snow, or ice pellets with moderate size drops. Precipitation seldom reaches the ground if the cloud base is above 10,000 feet.
- Altocumulus clouds are typically not associated with precipitation.

Precipitation Enhancers

Precipitation enhancers are substances or processes that increase the likelihood of precipitation. The most common enhancers include:

- Moist air advection increases relative humidity.
- Cold air advection decreases the atmosphere's ability to hold moisture thereby increasing relative humidity. This occurs the most behind cold fronts or during nighttime cooling.
- Diffluent or divergent flow aloft increases upward vertical motion bringing warmer, moist air to colder air aloft.

Visibility Forecast Techniques

Fog

Fog is a "cloud" forming below 50 feet and usually forms on the surface but can form aloft then build down to the surface. It is the most common visibility restrictor which can reduce visibility down to zero. Dissipation of fog can occur any one of three ways: remove or decrease moisture, heat the atmosphere, or change the surface wind speed.

Formation

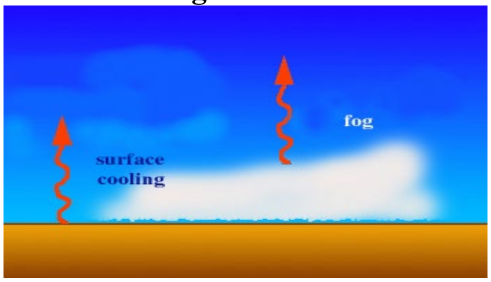
Fog needs three things for formation and all three must be present or fog never develops.

- *Increased Moisture*. You must have an abundant amount of moisture for fog to form.
- *Atmospheric Cooling*. The air needs to cool so relative humidity increases toward 100% or higher, forcing condensation to occur.
- *Turbulent Mixing*. For fog formation turbulent mixing must be present between the surface and boundary layer. Wind speeds must be high enough to increase moisture transport and support cold air advection, and they must be low enough to aid in fog formation. If the turbulent mixing is too much or too strong it will be difficult to achieve saturation due to mixing drier ambient air. Conversely if it is too low or too weak moisture remains on the surface forming dew or frost depending on surface temperature.

Radiation Fog

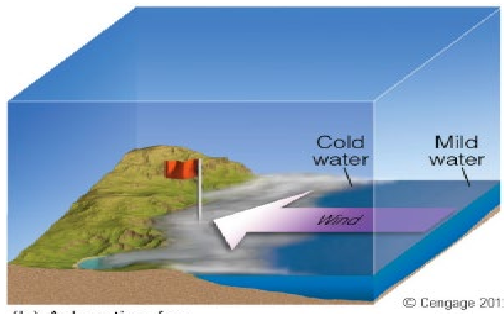
The first item needed is moist air trapped below a radiation inversion. Moisture may be from surface water sources like a lake or river, residual water from precipitation, or higher levels of relative humidity due to moist air advection. The second requirement is surface winds between 3-7KT. Clear skies overnight enhance radiation fog since it maximizes nighttime cooling.

Radiation Fog



Advection Fog

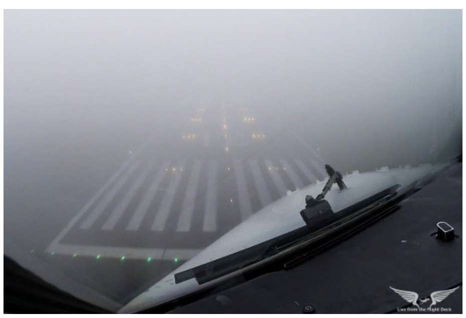
When warm, moist air moves over colder surfaces, the temperature drops due to diabatic cooling. This occurs most near lake or large river shorelines or coastal waters. It can occur inland as well where warm or moist air advection flows over continental or modified continental polar air. In addition to direction, surface wind speeds are also a factor. When winds are less than 3 knots, the advection fog tends to be shallow and if the speed is between 3 and 9 knots it is deep, dense fog.



(b) Advection fog

Frontal Fog

The three requirements for frontal fog formation are moist air forced aloft by a frontal boundary, evaporative cooling, and light surface winds. This can occur in two locations producing two types of frontal fog. Frontal fogs rarely dissipate. In most cases they simply move downstream with the surface fronts.

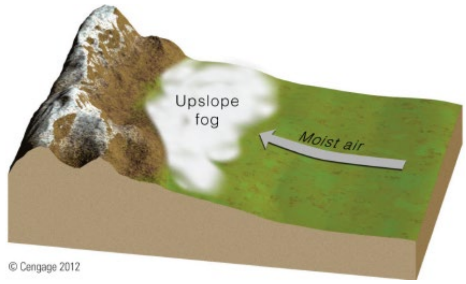


Pre-frontal Fog. Pre-frontal or warm-frontal fog is the most common type. It is a widespread area located in the cold air ahead of warm fronts often underneath a baroclinic leaf or comma cloud on satellite imagery.

Post-frontal fog. Post-frontal or cold frontal fog occurs less often since it is behind surface cold fronts. Since it requires evaporative cooling, it make sense that it occurs most often 150 to 250 miles behind active cold fronts. The exact location is between the surface front and 850-mb upper cold front.

Upslope Fog

Upslope fog forms when surface winds force moist air aloft by orographic lift. The winds perpendicular component to the higher terrain must have speeds less than 10KT. The closer the wind direction is to perpendicular increases the potential of upslope fog formation. Initially the fog forms at higher elevations then builds downward. If the condition extends over time, upslope fog may change to upslope stratus.



Blowing Dust or Sand

Blowing dust or sand is a function of wind speed and direction as well as unprotected, loose soil or sand. Essentially blowing dust or sand can occur with any weather feature creating low-level confluent or convergence upstream of an area with loose soil or sand.
Areas with wind speed greater than or equal to 15 knots are needed. Lower speeds rarely lift the soil or sand above six feet AGL. Additionally, the surface dew-point depression may be greater than or equal to 10°C.

Blowing Snow

The requirements for blowing snow are essentially the same as those for blowing dust or sand. The main difference is loose snow is needed, preferably freshly fallen snow. Winds must be strong enough to lift the snow above six feet AGL. 10 to 15 knots will lift loose, dry snow off the ground but to lift moderate, dry, fluffy snow wind speed must be 15 knots or higher.

Exercise / Practice: Precipitation & Visibility Forecasting

Temperature Forecast Techniques

No Airmass Change

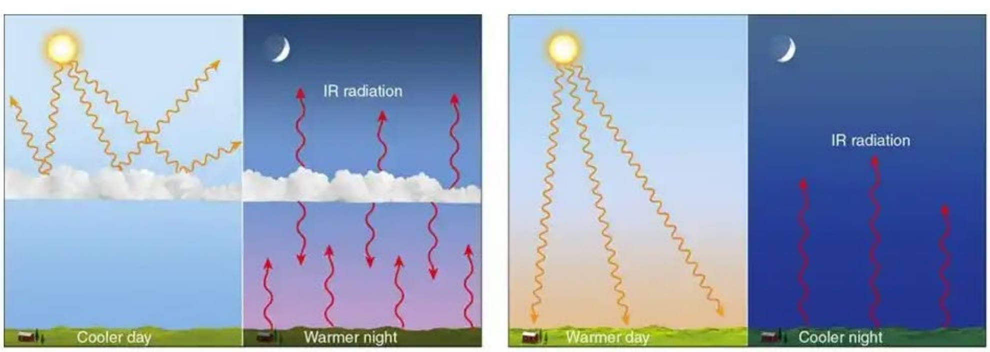
When the current air mass over a location does not change, forecast the temperatures to approximate the previous day's temperatures. Obviously, you should adjust for surface thermal advection, but diurnal effects usually dictate the trend of temperature changes the most. For example:

- The maximum temperature occurs in the mid-afternoon between 1400 and 1600.
- Minimum temperatures occur approximately one hour before or after sunrise.
- Temperatures increase the most during the morning hours and decrease the most after midnight.

METAR KTPA 201953Z 26007KT 240V310 10SM CLR 28/18 A3007 RMK AO2 SLP182 T02780178
METAR KTPA 211953Z 25007KT 10SM FEW035 28/17 A3006 RMK AO2 SLP180 T02780172

Cloud Coverage. You must always consider cloud coverage when forecasting temperatures. The amount of insolation producing surface heating can alter the forecast from one day to the next. If the cloud coverage stays constant from one day to the next, forecast the maximum and minimum temperatures to stay nearly the same. When low cloud ceilings increase coverage, the temperature range decreases. The maximum temperature will be lower, and the minimum temperature will be higher than the previous day. When low cloud ceilings decrease coverage, the temperature range increases. Now both the maximum and minimum temperatures will be lower than the previous day's temperatures.

Cloud Cover VS. Clear Skies

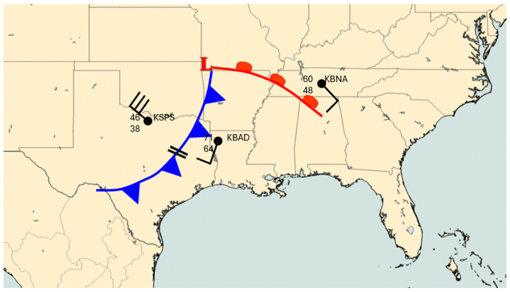


(a) Small daily temperature range (b) Large daily temperature range

https://www.memphisweather.blog/2020/03/how-long-will-wet-pattern-continue-and.html

Airmass Changes

Frontal Passage. After frontal passage temperatures change based on the frontal system that passed through your forecast location. Temperatures increase immediately after the passage of warm front. Cold fronts are different although temperatures cool gradually after their passage. The coldest temperature after an inactive cold front occurs an average of 12 to 24 hours after passage, while the coldest temperatures behind an active cold front occur an average of 24 to 48 hours after passage.

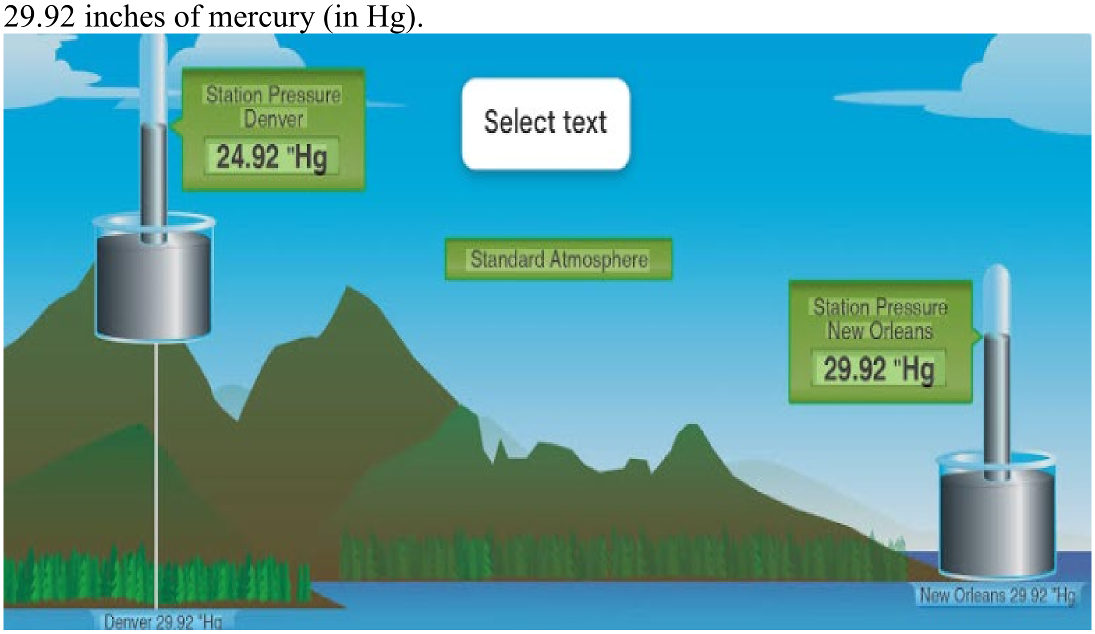


Thermal Advection. Thermal advection will modify temperatures whether the air mass changes or whether it stays the same. With surface warm air advection, the boundary layer destabilizes, temperatures increase, and cumuliform clouds may form. Cold air advection is the opposite. The boundary layer stabilizes, temperatures decrease, and stratiform clouds with possible and light precipitation may develop.

Pressure Forecast Techniques

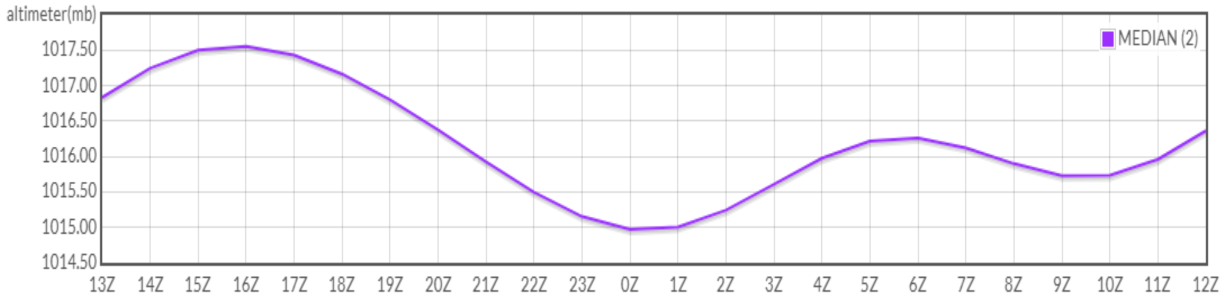
Surface Pressure

Surface pressure is the atmospheric pressure at a specific location. It is directly proportional to the air mass above a location. You know the global standard for sea-level pressure is 1013 mb or 29.92 inches of mercury (in Hg).



Diurnal Pressure Cycle

If there is no change in airmass, pressure rises and falls diurnally. Lower pressure is directly proportional to daytime heating, and higher pressure is directly proportional to nighttime cooling. The pressure rarely exceeds 3 mb daily.

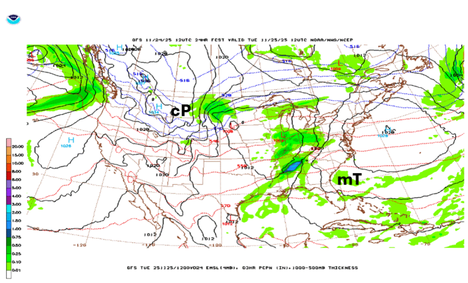


Migrating Weather Systems

To forecast pressure with migrating weather systems, you must first forecast the movement and intensity change in the approaching pressure center and modify the isobaric pattern due to terrain influences. When a lower pressure system approaches, you should forecast lower pressure and consider increasing wind speeds with an increasing isobar gradient, increasing temperatures with southerly winds, decreasing stability, and increasing cloud coverage with precipitation possible.

When a high-pressure system approaches, forecast increasing pressure and stability with decreasing wind speeds associated with a loosening isobar gradient. Any considerations after that depend on the air mass characteristics associated with the higher-pressure system.

Temperatures will decrease with a polar air mass after the cold front passes, or they will increase as maritime tropical air replaces retreating cold air. With a polar air mass expect fog, stratus, or stratocumulus to form after cold frontal passage then eventually decrease to clear skies.



A maritime tropical air mass brings an increased chance of cumuliform cloud development or an increased subsidence inversion that may trap pollutants near surface bringing the visibility down due to haze or smoke.

All pressure forecast techniques compute or determine pressure forecasts using sea-level pressure in millibars.

Exercise / Practice: Temperature & Pressure Forecast

Practice: Basic Forecasting Techniques
PC: Basic Forecasting Techniques